

EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?
2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?
3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?
4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?

IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. Open the Tool and Connect Dataset

- Open Tableau Desktop , Power BI Desktop.
- Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
- Preview the dataset to understand its structure.

2. Initial Inspection

- Explore fields such as Sales, Profit, Region, or Product Category.
- Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.

3. Handling Missing Values

- Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls.
- Replace missing numerical values with mean/median, and text with "Unknown".
- Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.

4. Detecting and Managing Outliers

- Create scatter or box plots for Sales and Profit to identify extreme values.
- Choose strategies:
 - Filter them temporarily.
 - Apply caps using calculated columns.

- Flag them with labels for review.

5. Resolving Inconsistencies

- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
- Normalize data formats, such as date fields or currency units.

6. Validation with Visuals

- Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
- Compare before/after cleaning stages for accuracy.

7. Save the Cleaned Dataset

- Save as .twbx (Tableau) / .pbix (Power BI).
- Optionally export cleaned data to .csv.

8. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).

Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

1. Sales
2. Profit
3. Region
4. Category

5. Order Date / Transaction Date

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?
2. How were outliers handled in your dataset and what visual confirmed their correction?
3. What inconsistencies were detected in categorical values and how were they standardized?
4. Describe how calculated fields helped in resolving data issues.
5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		